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FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2024

FUNDAMENTALS of MECHATRONICS

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**5 x2=10**		
1.	**a.**	Differentiate between an active and a passive transducer.	L2	1	**02**
	b.	List two key differences between microprocessors and microcontrollers.	L2	2	**02**
	c.	What is the function of the Program Counter in a microprocessor?	L2	3	**02**
	d.	What are two important factors to consider when selecting a motor for an actuation system?	L2	4	**02**
	e.	What is the main function of a directional control valve (DCV) in a hydraulic system?	L2	5	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	Explain the evolution of mechatronics, highlighting key technological advancements that contributed to its development.	L2	1	**05**
		OR			
	b.	Explain the working principle of photodiodes and their applications in mechatronic systems.	L2	1	**05**
3.	**a.**	Explain the basic architecture of a microprocessor system, including the CPU, memory, and I/O components.	L2	2	**05**
		OR			
	b.	Describe the functionS and importance of registers in microprocessor architecture.	L2	2	**05**
4.	**a.**	Compare and contrast CISC and RISC architectures in microprocessors.	L4	6	**05**
		OR			
	b.	Describe the interrupt system in the 8085 microprocessor, including the types of interrupts and their priorities.	L2	2	**05**
5.	**a.**	Discuss the concept of a latching circuit in PLC programming.	L2	3	**05**
		OR			
	b.	Explain some key factors to consider when choosing a PLC.	L2	3	**5**

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6.	a.	What is the concept of a "duty cycle" and how does it influence motor selection?	L2	4	5
		OR			
	b.	Explain the concept of gear trains and their role in mechanical actuation. Describe different types of gears and how they are used to transmit power and change speed or direction of rotation.	L2	4	5
7.	a.	Explain the principle of operation of a 4-way, 3-position, solenoid-operated sliding spool valve.	L2	5	5
		OR			
	b.	Describe the working principle of pressure relief valves and their importance in protecting pneumatic and hydraulic systems from excessive pressure.	L2	5	5
		PART-C	**6 X 10=60**		
8.	a.	Modern washing machines are a prime example of Mechatronics applications. Explain how different Mechatronics components work together to achieve various washing cycles and functions in an automatic washing machine.	L2	1	10
		OR			
	b.	Describe the working principle of light sensors, explaining how they convert light into an electrical signal.	L2	1	10
9.	a.	Imagine you are designing a temperature control system for a greenhouse. Explain how a microcontroller can be used to monitor temperature and activate actuators (e.g., fans, heaters) to maintain a desired temperature range	L2	2	10
		OR			
	b.	Describe the basic elements of a control system and explain how a microprocessor or microcontroller can be used to implement control functions in a Mechatronics system.	L2	2	10
10.	a.	Design the ladder logic for controlling the running state of the single phase motor by pressing START and STOP pushbuttons i.e. motor should remain in ON state after START pushbutton is pressed and should OFF when STOP pushbutton is pressed.	L3	3	10
		OR			

	b.	Explain the concept of "advanced actuators" in Mechatronics. Explain the working principle of any one of them.	L2	3	**10**
11.	**a.**	Discuss the functional requirements for an industrial robot in terms of accuracy, repeatability, and payload capacity.	L2	5	**10**
		OR			
	b.	What are end Effectors? Discuss on the different types of end effectors in robotics.	L2	5	**10**
12.	**a.**	Compare and contrast mechanical and electrical actuation systems.	L4	4	**10**
		OR			
	b.	Discuss the working principle of cams and their applications in mechanical actuation systems. Provide examples of how cams are used to achieve specific motion profiles.	L3	4	**10**
13.	**a.**	Classify direction control valves used in pneumatic and hydraulic systems based on their function. Explain how each valve type contributes to system operation.	L2	5	**10**
		OR			
	b.	Describe the key components of a hydraulic system and explain their individual functions.	L2	5	**10**